

iGEM DryLab Bonus 2

Mechanistic ODE Modeling of Antigen Expression Kinetics for an mRNA–LNP Vaccine Platform

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Abstract

This report develops a simple ordinary differential equation model for antigen expression in an mRNA–LNP vaccine platform after intramuscular administration. The model links uptake, endosomal escape, mRNA degradation, translation, and protein clearance to antigen expression over time. Simulations show that, even at the same dose, changes in trafficking and stability parameters can produce either sharp transient expression or lower but longer-lasting expression. Overall, the model offers a biologically grounded framework for comparing vaccine designs and analyzing antigen kinetics.

1 Background

The development of mRNA-LNP vaccines relies on a complex intracellular delivery cascade that translates a genetic sequence into a functional immunological stimulus. A significant hurdle in the field is the high variability in antigen expression profiles despite uniform dosing. This discrepancy stems from the sensitive dependence of expression kinetics on the physicochemical properties of the lipid carrier and the molecular design of the mRNA transcript. [1]

The problem is abstracted as a series of coupled kinetic hurdles: cellular uptake, endosomal escape—where typically only 1–3% of the payload successfully reaches the cytosol [2]—and the subsequent balance between translation and degradation. Understanding these temporal dynamics is essential because effective immune activation is a threshold-dependent process, requiring specific antigen concentrations and presentation durations to trigger robust B-cell and T-cell responses. This modeling effort provides a mechanistic framework to explore how individual biological parameters shape the sharp versus sustained profiles observed in early experimental results, facilitating the rational design of more potent vaccine platforms.

2 Model Structure and Biological Assumptions

Modeling framework

We use a **compartmental ODE model** (linear, first-order kinetics) describing a chain:

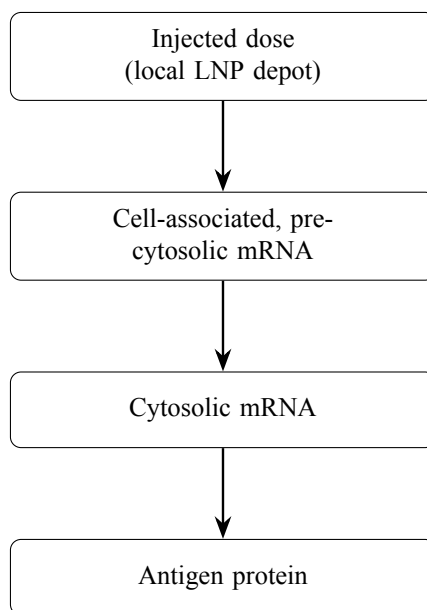


Figure 1: Simplified mRNA vaccine expression pipeline.

This explicitly represents the biological sequence requested:

1. **Delivery** (availability + cellular uptake)
2. **Release & degradation** (productive cytosolic release vs. nonproductive loss; cytosolic decay)
3. **Translation** (protein synthesis from cytosolic mRNA)
4. **Antigen expression** (protein accumulation with clearance)

5. **Immune activation** (modeled via interpretable exposure metrics and, optionally, a minimal “innate brake”)

State variables

All state variables are **lumped, tissue-level “effective amounts”** within the relevant expressing compartment (e.g., injection site + draining immune microenvironment). Units can be mass (e.g., μg equivalents) or arbitrary units (AU) depending on available measurements.

- $L(t)$: extracellular/interstitial **LNP-associated mRNA depot** that remains available for cellular uptake at the site (or local microenvironment).
- $E(t)$: **cell-associated, pre-cytosolic mRNA** (internalized/endosomal/vesicular; not yet available for ribosomal translation).
- $M(t)$: **cytosolic mRNA** available for translation.
- $P(t)$: **antigen protein** (intracellular + surface-associated).

Optional immune-related variable (used only when needed to reproduce “peak then crash” behavior):

- $I(t)$: a dimensionless **translation-inhibition/innate-stress** proxy induced by RNA/LNP sensing that downmodulates translation.

Core assumptions

1. **First-order kinetics** for uptake, release, and degradation. This is a standard approximation when detailed molecular steps are aggregated and when concentrations are below strong saturation regimes. [1]
2. **Well-mixed effective compartment**: spatial biodistribution and cell-to-cell variability are not explicitly modeled (limitations discussed later).
3. **Two competing fates** for internalized mRNA: productive cytosolic release vs. nonproductive loss. The low cytosolic delivery efficiency widely reported for LNP systems motivates keeping this competition explicit. [2]
4. **Translation proportional to $M(t)$** in the main model (linear translation). Saturation can be added later if data show plateaus inconsistent with linearity; in vitro evidence of saturation with LNP mRNA transfection has been noted. [1]
5. **Protein clearance is first-order** (net degradation + secretion + cell turnover lumped).

Despite simplification, the model encodes the most influential “knobs” known to shift expression kinetics — uptake/depot persistence, endosomal escape, mRNA stability, translation potency, and protein turnover. Therefore, the it can be considered biologically reasonable.

3 Mathematical Formulation

Main model: 4-state

Multi-day translation signatures at injection sites have been observed, suggesting prolonged availability and/or ongoing uptake in some contexts. [1] Therefore, it is valuable to include a **local depot + uptake** step explicitly

upon basic intracellular model. We hence choose the following 4-state model as the main model:

$$\frac{dL}{dt} = -(k_{\text{up}} + k_{\text{clr},L})L, \quad (1)$$

$$\frac{dE}{dt} = k_{\text{up}}L - (k_{\text{rel}} + k_{\text{loss},E})E, \quad (2)$$

$$\frac{dM}{dt} = k_{\text{rel}}E - k_{\text{deg}}M, \quad (3)$$

$$\frac{dP}{dt} = k_{\text{tl}}M - k_{\text{clr}}P. \quad (4)$$

Meaning of each term

- $k_{\text{up}}L$: **delivery into cells** (effective uptake rate of depot cargo into a cell-associated compartment).
- $k_{\text{clr},L}L$: **loss of depot availability** (drainage, extracellular degradation, irreversible dispersion).
- $k_{\text{rel}}E$: **productive endosomal escape / cytosolic release** (functionally available mRNA).
- $k_{\text{loss},E}E$: **nonproductive loss of internalized mRNA** (endosomal degradation, recycling/ exocytosis, irreversible trapping). Low net cytosolic delivery motivates allowing $k_{\text{loss},E} \gg k_{\text{rel}}$ in some regimes. [2]
- $k_{\text{deg}}M$: **cytosolic mRNA decay** (RNases, decay pathways). Median mammalian mRNA half-lives on the order of hours support k_{deg} in the $0.03 - 0.3 \text{ h}^{-1}$ range, though delivered IVT mRNA can differ substantially. [3]
- $k_{\text{tl}}M$: **translation rate** (effective protein production per cytosolic mRNA).
- $k_{\text{clr}}P$: **net antigen clearance/degradation**. Protein half-lives span wide ranges (hours to days depending on state/tissue), supporting a broad plausible range for k_{clr} . [4]

Optional extension: minimal innate “translation brake” (when sharp drops occur)

To capture cases where expression peaks and then rapidly collapses via innate responses (e.g., strong IFN-associated translation block), add:

$$\frac{dI}{dt} = k_I M - k_{I,\text{decay}} I \quad (5)$$

and replace the translation term with an inhibited form:

$$\frac{dP}{dt} = k_{\text{tl}} \frac{M}{1 + I} - k_{\text{clr}}P. \quad (6)$$

Rationale: innate sensing of RNA/LNP can lead to stress and reduced translation; such feedback is described mechanistically and observed experimentally, including profiles consistent with a rapid innate response followed by decreased expression. [5]

Initial conditions

For a single bolus injection at $t = 0$:

$$L(0) = \eta_{\text{avail}} D_0, \quad E(0) = 0, \quad M(0) = 0, \quad P(0) = 0. \quad (7)$$

- D_0 : injected mRNA dose (mass or AU).
- η_{avail} : fraction of dose that remains as a **bioavailable local depot** (captures losses during injection, aggregation, immediate clearance). If unknown, fold it into D_0 .

Parameters

The numerical values below are **illustrative** (not fitted). They are chosen to generate a peak on the order of ~ 1 day with multi-day tails, roughly consistent with reported in vivo antigen/reporter kinetics after intramuscular administration. [6]

Key external anchors used qualitatively:

- Cytosolic delivery efficiency often $\sim 1\text{--}3\%$ of internalized cargo. [2]
- Protein expression peaks can occur within hours to ~ 1 day depending on route/tissue, with translation signals detectable for days in some contexts. [1]
- Median mammalian mRNA half-life estimates on the order of hours (context-dependent). [3]
- Protein half-lives in mammalian systems span hours to days, depending on cell state/tissue. [4]

Baseline example (used for scenario comparisons in a later section)

Time in hours:

$$\begin{aligned}
 D_0 &= 1 \quad (\text{normalized}), \\
 \eta_{\text{avail}} &= 1 \quad (\text{absorbed into } D_0), \\
 k_{\text{up}} &= 0.10 \text{ h}^{-1}, \\
 k_{\text{clr},L} &= 0.02 \text{ h}^{-1}, \\
 k_{\text{rel}} &= 0.20 \text{ h}^{-1}, \\
 k_{\text{loss},E} &= 9.80 \text{ h}^{-1} \quad (\text{gives escape fraction } \approx 0.02 [2]), \\
 k_{\text{deg}} &= 0.10 \text{ h}^{-1}, \\
 k_{\text{tl}} &= 1.0 \text{ (protein AU)(mRNA AU)}^{-1}\text{h}^{-1}, \\
 k_{\text{clr}} &= 0.03 \text{ h}^{-1}.
 \end{aligned}$$

4 Expected Dynamic Behavior

Qualitative shape of $P(t)$

Even without the innate brake, the 4-state chain produces a characteristic delayed rise $\rightarrow$ peak $\rightarrow$ decay, because antigen production requires sequential progression through multiple steps (depot $\rightarrow$ uptake $\rightarrow$ cytosol $\rightarrow$ protein). This is consistent with many reported mRNA–LNP expression profiles where maximal signal occurs after several hours and then declines. [1]

The model also naturally produces:

- **Delay before antigen accumulation:** controlled mainly by $1/(k_{\text{up}} + k_{\text{clr},L})$ and $1/(k_{\text{rel}} + k_{\text{loss},E})$.

- **Peak timing:** controlled by the slowest upstream rates plus protein turnover.
- **Persistence/tail:** controlled by the slower of (i) depot persistence, (ii) cytosolic mRNA decay, and (iii) protein clearance.

Adding the optional $I(t)$ module can yield sharp drops after an early peak, consistent with cases where strong innate responses produce translational inhibition. [7]

Analytical insights

Because the main model is linear, several “global” outcomes can be derived without simulation.

Escape and uptake fractions

Define two dimensionless efficiencies:

$$f_{\text{up}} = \frac{k_{\text{up}}}{k_{\text{up}} + k_{\text{clr},L}} \quad (\text{fraction of depot that becomes internalized}), \quad (8)$$

$$f_{\text{esc}} = \frac{k_{\text{rel}}}{k_{\text{rel}} + k_{\text{loss},E}} \quad (\text{fraction of internalized mRNA that becomes cytosolic}). \quad (9)$$

Reported cytosolic delivery efficiencies for LNP systems motivate $f_{\text{esc}} \ll 1$ in many regimes (order 0.01–0.03). [2]

Total cytosolic mRNA exposure

Integrate $dM/dt = k_{\text{rel}}E - k_{\text{deg}}M$ from 0 to ∞ (with $M(0) = M(\infty) = 0$):

$$\int_0^\infty k_{\text{rel}}E(t) dt = k_{\text{deg}} \int_0^\infty M(t) dt. \quad (10)$$

Similarly, conservation through the upstream chain gives total cytosolic “arrivals”:

$$\int_0^\infty k_{\text{rel}}E(t) dt = D_0 f_{\text{up}} f_{\text{esc}}. \quad (11)$$

Therefore,

$$\int_0^\infty M(t) dt = \frac{D_0 f_{\text{up}} f_{\text{esc}}}{k_{\text{deg}}}. \quad (12)$$

Interpretation: cytosolic mRNA exposure scales multiplicatively with (i) how much dose is internalized, (ii) how much escapes, and (iii) the cytosolic lifetime.

Antigen exposure (AUC) as a product of mechanisms

Integrate $dP/dt = k_{\text{tl}}M - k_{\text{clr}}P$ (with $P(0) = P(\infty) = 0$):

$$k_{\text{tl}} \int_0^\infty M(t) dt = k_{\text{clr}} \int_0^\infty P(t) dt. \quad (13)$$

So:

$$\text{AUC}_P \equiv \int_0^\infty P(t) dt = D_0 f_{\text{up}} f_{\text{esc}} \frac{k_{\text{tl}}}{k_{\text{deg}} k_{\text{clr}}}. \quad (14)$$

Interpretation:

- If endosomal escape is 1–3%, it can dominate the overall scaling even if translation is strong. [2]
- Improvements in mRNA chemistry/purity that reduce innate activation can increase the effective k_{tl} by preventing translation shutdown, linking immunogenicity to expression kinetics. [5]

Peak vs persistence trade-off

In this linear chain, higher persistence can be achieved by reducing $k_{\text{clr},L}$, k_{deg} , or k_{clr} , but those changes often shift the peak later and broaden the curve (longer transit/accumulation). Conversely, aggressive early uptake/release (large k_{up} , effectively larger early flux) can create early high peaks but short duration if clearance/decay rates are also high.

5 Simulation Plan and Example Scenarios

Simulation workflow

1. Implement the ODEs for (L, E, M, P) and solve over $t \in [0, T]$ (e.g., 0–10 days) using a stiff-capable solver if $k_{\text{loss},E}$ is large (common when modeling low escape).
2. Compute interpretable summary metrics from $P(t)$:
 - P_{max} (peak height)
 - t_{peak} (time to peak)
 - AUC_P (total antigen exposure)
 - FWHM (full width at half maximum; duration near peak)
 - time-above-threshold $T(P \geq P_{\text{thr}})$ (immune-relevant persistence surrogate)
3. Perform scenario sweeps and local sensitivities on log-scaled parameters.

Example baseline and scenario outcomes

Using the baseline parameter set defined earlier (normalized dose), a simple numerical integration yields a single-peaked antigen curve with a peak near ~ 1 day and multi-day tail.

Below are model-generated metrics for several contrasting scenarios (all else held constant unless stated). These numbers are model outputs for illustration, not measured data.

Table 1: Illustrative scenario comparison

Scenario (change vs baseline)	Peak $P_{\max}$	t_{peak} (h)	AUC _P	FWHM (h)
Baseline	0.089	27.8	5.55	54.2
Fast mRNA decay $k_{\text{deg}} = 0.30$	0.034	19.7	1.85	45.5
Slow mRNA decay $k_{\text{deg}} = 0.03$	0.197	43.7	18.37	84.7
Low translation $k_{\text{tl}} = 0.3$	0.027	27.8	1.66	54.2
High translation $k_{\text{tl}} = 3.0$	0.267	27.8	16.64	54.2
Poor delivery (slow uptake) $k_{\text{up}} = 0.03$	0.051	38.0	3.99	70.0
Efficient delivery (fast uptake) $k_{\text{up}} = 0.30$	0.110	21.1	6.24	48.2
Lower escape $f_{\text{esc}} = 0.01$	0.044	27.7	2.77	54.2
Higher escape $f_{\text{esc}} = 0.10$	0.444	28.2	27.74	54.2
Fast protein clearance $k_{\text{clr}} = 0.10$	0.048	18.9	1.67	32.0
Slow protein clearance $k_{\text{clr}} = 0.01$	0.123	37.7	14.83	106.8

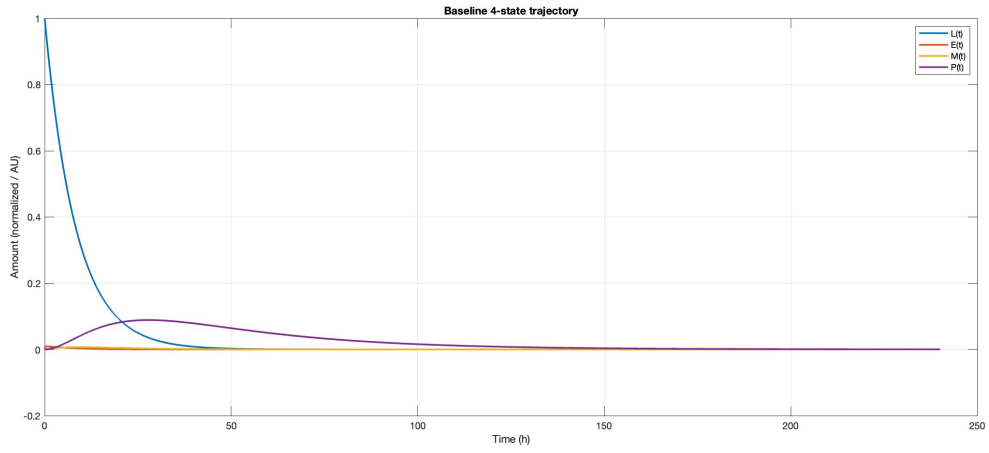


Figure 2: Baesline 4-state trajectory

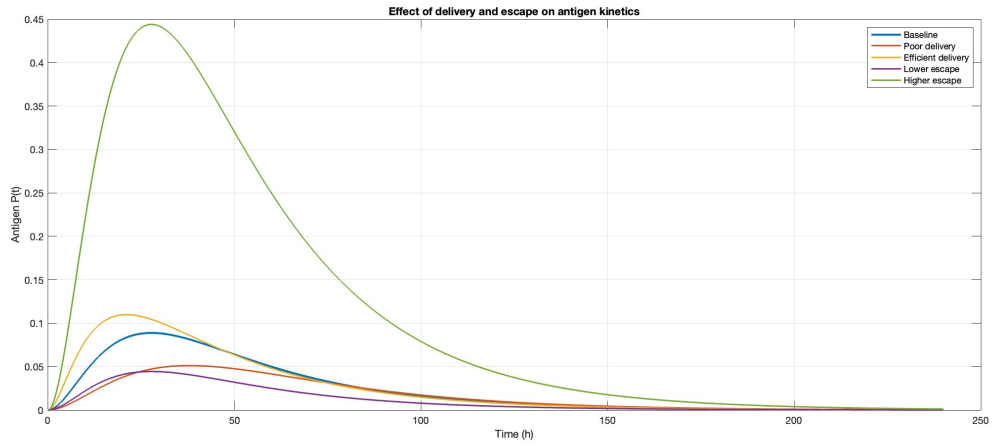


Figure 3: Effects of delivery and escape on antigen expression

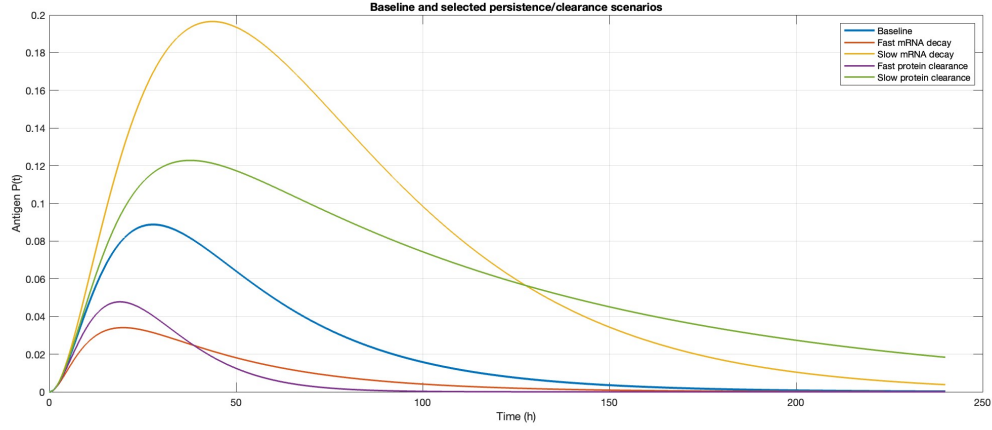


Figure 4: Baseline and selected scenarios

How these shifts map to the requested interpretation:

- **Peak height** increases strongly with k_{tl} , f_{esc} , and (in many regimes) faster early supply via k_{up} .
- **Time to peak** shifts earlier with faster uptake and faster clearance; shifts later with slower mRNA decay and slower protein clearance.
- **Duration** (FWHM / long tail) lengthens with slower mRNA decay and slower protein clearance; it also lengthens when uptake is slow but persistent (depot-limited regime).
- AUC_P follows the derived multiplicative scaling $D_0 f_{up} f_{esc} (k_{tl} / (k_{deg} k_{clr}))$, so changing f_{esc} by an order of magnitude changes AUC by an order of magnitude.

Burst vs sustained delivery strategies

Two stylized strategies implemented by changing depot/uptake parameters:

- **Burst-like**: rapid uptake and rapid depot loss (e.g., large k_{up} , $k_{clr,L}$).
- **Sustained-like**: slower uptake with lower depot clearance (smaller k_{up} , $k_{clr,L}$).

Illustrative outputs:

- Burst: earlier peak (~ 20 h), modestly narrower curve.
- Sustained: later peak (~ 43 h), broader curve with longer tail.

Such differences are consistent with observations that expression kinetics depend on local stability/availability and trafficking, not only total dose. [1]

Matlab simulation psuedocode

```

1 Algorithm: Simulation and metric extraction for the mRNA vaccine model
2
3 Input:
4   Baseline parameters  $\theta = \{D_0, \eta_{avail}, k_{up}, k_{clr,L}, k_{rel}, k_{loss,E}, k_{deg}, k_{tl},$ 
5      $\hookrightarrow k_{clr}\}$ 
6   Simulation time interval  $[0, T]$ 
7   Scenario set S
8
9 For each scenario s in S do

```

```

9      1. Replace baseline parameters  $\theta$  by the scenario-specific values  $\theta_s$ 
10
11     2. Initialize state variables:
12          $L(0) = \eta_{\text{avail}} \cdot D_0$ 
13          $E(0) = 0$ 
14          $M(0) = 0$ 
15          $P(0) = 0$ 
16
17     3. Define the ODE system:
18          $dL/dt = -(k_{\text{up}} + k_{\text{clr},L}) \cdot L$ 
19          $dE/dt = k_{\text{up}} \cdot L - (k_{\text{rel}} + k_{\text{loss},E}) \cdot E$ 
20          $dM/dt = k_{\text{rel}} \cdot E - k_{\text{deg}} \cdot M$ 
21          $dP/dt = k_{\text{tl}} \cdot M - k_{\text{clr}} \cdot P$ 
22
23     4. Numerically solve the ODE system on  $[0, T]$ 
24
25     5. From the simulated antigen trajectory  $P(t)$ , compute:
26          $P_{\text{max}} = \max P(t)$ 
27          $t_{\text{peak}} = \text{time at which } P(t) \text{ reaches } P_{\text{max}}$ 
28          $\text{AUCP} = \text{integral of } P(t) \text{ over time}$ 
29          $\text{FWHM} = \text{duration for which } P(t) \geq P_{\text{max}} / 2$ 
30
31     6. Store the metric results of scenario  $s$ 
32 End
33
34 Output:
35     Time-course curves of  $L(t)$ ,  $E(t)$ ,  $M(t)$ ,  $P(t)$ 
36     Summary table of  $P_{\text{max}}$ ,  $t_{\text{peak}}$ ,  $\text{AUCP}$ , and  $\text{FWHM}$  for all scenarios

```

6 Biological Interpretation

Mechanisms producing a sharp, short-lived peak

In this model, sharp transient expression arises when antigen production is front-loaded and/or clearance is fast:

1. **Fast uptake + fast depot loss** (k_{up} high, $k_{\text{clr},L}$ high): many cells get payload quickly, but the source is exhausted rapidly $\rightarrow$ early peak, limited persistence.
2. **Fast cytosolic mRNA decay** (k_{deg} high): even if delivery is good, mRNA disappears quickly $\rightarrow$ short translation window.
3. **Fast antigen clearance** (k_{clr} high): protein cannot accumulate; peaks earlier and decays quickly.
4. **Innate-driven translation shutdown (optional module)**: strong innate sensing can yield a peak followed by a sharp drop consistent with translation blockage. This mechanism is supported by reports where LNP-delivered RNA induced rapid IFN responses and was associated with abrupt expression loss. [7]

Mechanistically, innate sensing can be triggered by both (a) the RNA cargo (especially unmodified uridine or dsRNA impurities) and (b) the LNP components, and downstream signaling can reduce translation. [5]

Mechanisms producing lower but longer-lasting expression

Sustained expression emerges when the input to translation is time-distributed and/or the degradation/clearance rates are slower:

1. **Prolonged depot availability** (smaller $k_{\text{clr},L}$) and/or **slow uptake** (smaller k_{up}): extends the time over which mRNA becomes cytosolic, lowering peak but lengthening duration.
2. **Improved cytosolic mRNA stability** (smaller k_{deg}): increases both AUC and duration; also tends to shift peak later because accumulation continues longer.
3. **Longer antigen half-life** (smaller k_{clr}): increases AUC and extends tails even if mRNA is transient.

Experimentally, multi-day translation signatures have been measured in some tissues/routes, and antigen protein expression can persist beyond the period where intact mRNA is abundant — consistent with protein half-life effects and time-distributed delivery. [1]

Which parameters are most influential?

The analytical AUC result highlights multiplicative bottlenecks:

- $f_{\text{esc}} = k_{\text{rel}}/(k_{\text{rel}} + k_{\text{loss},E})$ can be very small (order 0.01–0.03), so improving it can produce large gains.
- $f_{\text{up}} = k_{\text{up}}/(k_{\text{up}} + k_{\text{clr},L})$ can vary widely with formulation and microenvironment; it strongly scales overall output.
- k_{deg} and k_{clr} control duration and AUC through $1/(k_{\text{deg}}k_{\text{clr}})$.
- k_{tl} is a potent scaling factor, but its effective value can be suppressed by innate activation.

Are “strong” and “long-lasting” synergistic or a trade-off?

Both effects can be synergistic in terms of AUC (e.g., increasing f_{esc} and lowering k_{deg} directly multiply). However, peak height vs. duration can trade off:

- Slowing upstream delivery (smaller k_{up} , smaller $k_{\text{clr},L}$) tends to reduce peak while increasing duration (a classic depot effect).
- Increasing protein stability (smaller k_{clr}) can raise both peak and duration, but may shift peak later and prolong exposure beyond desired windows (which may affect reactogenicity or tolerance risk; outside scope of this simple model).
- Strong innate responses may simultaneously improve adjuvanticity and reduce translation, creating a non-trivial trade-off between immune stimulation and antigen production.

7 Sensitivity and Design Insights

Sensitivity structure

For the linear main model, antigen exposure satisfies:

$$\text{AUC}_P \propto D_0 \times f_{\text{up}} \times f_{\text{esc}} \times \frac{k_{\text{tl}}}{k_{\text{deg}}k_{\text{clr}}}. \quad (15)$$

This implies:

- If f_{esc} is very small (as commonly reported), improving endosomal escape can dominate compared with modest changes in other parameters.
- If protein clearance is fast (high k_{clr}), improvements in mRNA stability may have diminishing returns on peak (protein cannot accumulate), though they still increase cumulative production.
- If innate activation reduces translation, then improving mRNA chemistry/purity can increase the effective k_{tl} and prevent collapse dynamics.

Design levers mapped to parameters

1. Increase fraction of cells receiving mRNA / effective delivery

- In model terms: increase η_{avail} and/or increase k_{up} relative to $k_{\text{clr},L}$.
- Empirically, trafficking and expression occur in specific immune and stromal cell populations at injection sites and draining lymph nodes, implying delivery can be highly cell-type- and tissue-structured. [6]

2. Improve endosomal escape / functional cytosolic delivery

- Increase f_{esc} by increasing k_{rel} and/or decreasing $k_{\text{loss},E}$.
- Endosomal escape is widely regarded as a primary bottleneck with low efficiencies reported for standard systems, motivating this as a high-impact lever. [2]

3. Tune release speed vs persistence

- Adjust k_{up} and $k_{\text{clr},L}$ to shape burst vs. sustained profiles.
- This is a direct mechanistic explanation for why two designs with identical dose can look different: the input flux to cytosolic mRNA is time-shaped.

4. Increase effective translation while controlling innate sensing

- Increase k_{tl} (e.g., UTR optimization, codon usage, chemistry) and/or reduce innate-driven inhibition (optional $I(t)$).
- Nucleoside modification and dsRNA impurity reduction have been shown to reduce innate immune activation and improve protein expression consistency. [5]

Calibration

A practical calibration strategy is to fit $P(t)$ (and ideally $M(t)$ proxies) from experiments such as:

- **Reporter bioluminescence time courses** (luciferase) to capture effective translation kinetics; luciferase enzymatic half-life can be short (hours), making it a closer proxy for active translation rather than long-lived protein accumulation.
- **Tissue mRNA quantification** over time (e.g., bDNA assays, in situ hybridization) and protein localization (IHC), which have been used to measure time courses in animal models and nonhuman primates, including peak mRNA early and peak protein later.

To fit the

1. Fit the linear model first (least squares / maximum likelihood) to obtain $(k_{\text{up}}, k_{\text{clr},L}, k_{\text{deg}}, k_{\text{clr}})$ identifiable

from timing/shape.

2. Then assess whether residuals show “peak then crash” behavior; if so, introduce the $I(t)$ module and fit $(k_I, k_{I,\text{decay}})$.

8 Limitations of the Model

This model is intentionally simplified; key omissions that can matter in real vaccines include:

- **Cell-to-cell heterogeneity:** distribution of LNP copies per cell and variability in translation capacity can broaden kinetics beyond a single-compartment average. [1]
- **Tissue distribution and pharmacokinetics:** biodistribution to liver/spleen/lymph nodes and time-varying trafficking alter both the magnitude and the location of antigen production; empirical data show tissue-specific time courses. [6]
- **Innate immune sensing complexity:** the optional $I(t)$ is a coarse proxy; real pathways involve endosomal TLRs, cytosolic sensors, cytokine cascades, and cell recruitment, and can differ across formulations and impurity profiles. [5]
- **Nonlinear translation saturation:** uptake/translation saturation has been observed in vitro, implying that at higher doses or high intracellular concentrations, linear $k_{tl}M$ may overpredict. [1]
- **Adaptive immune feedback:** antigen-specific killing of expressing cells, antibody-mediated clearance and lymphocyte dynamics can reshape later time courses; not represented.
- **Antigen processing/presentation details:** surface presentation, secretion, and MHC loading are collapsed into a single protein pool $P(t)$, whereas actual immune activation depends on antigen-presenting cell biology and context.

These limitations are acceptable for an initial platform model aimed at explaining why kinetics differ at equal dose, but they constrain direct quantitative prediction without calibration.

9 Final Conclusion

A small ODE model with an **LNP depot**, **cell-associated pre-cytosolic mRNA**, **cytosolic mRNA**, and **antigen protein** is sufficient to reproduce the key antigen-expression phenotypes observed in mRNA–LNP vaccination: delayed rise, a peak, and a decaying tail, with strong dependence on delivery and stability parameters.

The most interpretable mechanistic insight is that **total antigen exposure (AUC)** factorizes into a product of step efficiencies and lifetimes:

$$\text{AUC}_P \propto D_0 \times f_{\text{up}} \times f_{\text{esc}} \times \frac{k_{tl}}{k_{\text{deg}}k_{\text{clr}}}, \quad (16)$$

so small efficiencies — especially endosomal escape, widely reported to be low — can dominate platform performance.

Differences between sharp/transient vs. lower/sustained expression at fixed dose can be explained by:

- time-shaping of supply (burst vs. sustained depot/uptake),
- mRNA and protein lifetimes, and

- innate-response–driven translation suppression, which can be included as a minimal inhibitory state when abrupt post-peak drops are observed.

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